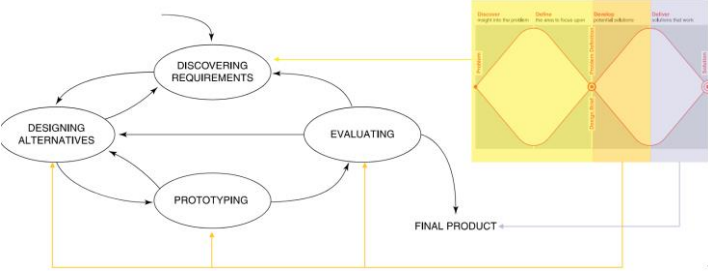


INTERACTION DESIGN

- **Definition:** the umbrella term covering disciplines fundamental, fields, approaches concerned with researching and designing computer-based system
- **Characteristics:**
 - 1) Users should be involved throughout the developments of project
 - 2) At beginning, specific usability, user experience goals should be identified and documented.
 - 3) Iteration is needed through the core activities
- **Importance of involving users:**
 - 1) **Understanding** of user goals leading better products
 - 2) **Expectation management** [realistic expectations, no surprises\ disappointments, time training, communication]
 - 3) **Ownership** [user→active stakeholder, more likely to forgive or accept problems, big difference in acceptance and success of product]
- **User experience (Subjective):** Product performance and usage in the real world. Designer should understand users, be aware of sensitivities and capabilities, (universality: no error assumption for disabilities)
- **Usability goals (Objective):** effectiveness (good), efficiency (easy/ fast), safety utility, learnability (用户能否通过探索界面学会使用产品), memorability (帮助用户记住如何完成任务)



- **User-entered approach Principles:**
 - 1) **Early focus on users and tasks**
 - Users' task and goals are the driving force behind the development
 - Users' behavior and context of use are studied, system is designed to support them
 - Users' characteristics are captured and designed for
 - Users' are consulted throughout development from earliest phases to the latest
 - All design decisions are taken within the context of the users (activities and environment)
 - 2) **Empirical measurement**
 - 3) **Iterative design**

DISCOVERING REQUIREMENTS

- **Describing Requirements:** a statement about an intended product that specifies what is expected to do or how it performs
 - 1) Atomic requirement shell
 - 2) User stories (as a [], I want to [], so that [])
 - 3) Types: functional requirement, data requirement, environment requirement, user characteristics, usability goals, user experience goals
 - 4) Requirement capturing interactions:
 - Personas
 - Scenario [textual descriptions, animations/ audio/ video, example animation scenarios]
 - Use cases:
 - ◆ Essential use case: division of tasks, no implementation detail
 - ◆ More alternative courses is more detail
 - 5) Gathering techniques:
 - Observation (direct / indirect)
 - Interviews (individual / group)
 - Daries, surveys, think-aloud evaluation, working prototype evaluation, studying documentation
 - Evaluating other system, ethnographic study, usability tests

- **Accessibility and Inclusiveness:** Disability [sensory impairment, physical impairment, cognitive] 种类包括 (permanent 永久的, temporary 暂时的, situational 嘈杂环境听不到)
- **Design Principles:**
 - 1) Visibility -- 方便找到想找的东西
 - 2) Feedback -- 对用户操作给予反馈
 - 3) Constraints -- 防止错误操作的操作
 - 4) Consistency -- similar operations and elements for similar tasks
 - 5) Internal Consistency -- 在 APP 里保持操作一致性
 - 6) External Consistency -- 不同 APP 保持一致性
 - 7) Affordances -- 物体的属性可以使人们知道如何使用 (符合 mental models)
- **Schneiderman's Eight Golden Rules:**
 - 1) Strive for **consistency**
 - 2) Enable frequent users to use **shortcuts**
 - 3) Offer informative **feedback**
 - 4) Design dialog to **yield closure**
 - 5) PROCOffer simple **error handling**
 - 6) Permit **easy traversal of actions**
 - 7) Support **internal locus of control**
 - 8) Reduce short-term **memory load**

ALTERNATIVE DESIGN

- **How to generate alternatives:**
 - 1) Stick with something works
 - 2) Consider alternatives helps identify better designs
- **Where alternatives come from:**

- 1) **Flair and creativity:** research and synthesis
- 2) Cross-fertilization of idea from different perspectives
- 3) Product evolution based on changing use
- 4) Seek inspiration: same products and domains, or different products and domain
- 5) Balancing constrains and trade-offs
- **How to choose among alternatives:**
 - 1) Interaction **design focus on externally**-visible and measurable behavior use of PC, CPU, memory
 - 2) Technical feasibility
 - 3) Evaluation with users and stakeholders: prototypes, not static documentation, because haviour is key
 - 4) A/B testing: inform the choice between alternatives
 - 5) Quality thresholds: think of the photo-taking experience

INTERACTION TYPES

- **Instruction:** 用户告诉系统要做什么 (一种常见的概念模型, 适用于各种设备和系统), 支持快速高效的交互, 适用于重复性操作
- **Conversing:** 模仿人与人之间对话, 语音识别、自然语言对话, 用户能够以熟悉的方式交互, 感到舒适放心, 但当系统无法解析用户说的话会导致 misunderstanding. 适用于查找信息或请求音乐
- **Manipulating:** 利用用户在物理世界中的移动和操作只是, 在虚拟对象上进行拖动、选择、打开、关闭、缩放。新手可以快速学习, 经验丰富的操作更快, 间歇性用户可以保留操作概念, 几乎不需要错误信息, 用户行动的即时性, 用户体验焦虑少, 可以获得信心和操纵感。但手势易出错, 不支持重复性任务, 鼠标等可能比光标慢, 有些人过于字面理解直接操作的隐喻。对于 **direct manipulation:** 适用于需要用户亲自执行操作的任务 (如设计、绘画、飞行、驾驶)
- **Exploring:** 用户在虚拟或物理环境中移动, 可嵌入传感器, 触发数字事件
- **Responding:** 系统主动提醒用户对其认为感兴趣的事, 适用于自动系统响应, 无需用户发出请求
- **Hybrid conceptual models** 混合概念模型: 适合支持**多种执行相同操作**的方式, **复杂的任务和用户群体**
- **Choosing an interaction type (怎么判断这个工作适合什么样的交互类型)**
- **Interaction and interface**
 - Interaction: A description of what the user is doing when interacting with a system (instructing, talking, browsing, or responding)
 - Interface: The kind of interface used to support the interaction (command, menu-based, gesture, or voice)

1. Command	11. Gesture
2. Graphical	12. Haptic
3. Multimedia	13. Multimodal
4. Virtual reality	14. Shareable
5. Web	15. Tangible
6. Mobile	16. Augmented reality
7. Appliance	17. Wearables
8. Voice	18. Robots and drones
9. Pen	19. Brain-computer interaction
10. Touch	20. Smart

Different kinds of interfaces

CONCEPTUALIZING DESIGN

- **Proof of concept:** Conceptualize what the proposed product will do
- **Necessary:** To **scrutinize vague ideas and assumptions** about the benefits of the proposed product in terms of their feasibility; How realistic is it to develop? How desirable and useful?
- **Assumption:** Taking something for granted when it needs further investigation
- **Claim:** stating something to be true when it is still open to question
- **Pros:**
 - Orientation: 帮助设计团队提出关于概念模型如何被理解的具体问题
 - Open-minded: 防止设计团队过早地聚焦于狭窄的视角
 - Common ground: 允许团队建立一套共同认可的艺术语
- **From problem space to design space**
 - Having a good understanding of the **problem space** can help inform the **design space** (For example, what kind of interface, behavior, functionality to provide)
 - Before deciding upon these, it is important to develop a **conceptual model**

CONCEPTUAL MODEL

- **Definition:** A **high-level description** of how a system is organized and operates; Provides **a working strategy** and framework of general concepts and their interrelations. (有效的概念模型看起来简单明了, 支持的操作直观易用)
- **Core Components:**
 - 1) Metaphors and analogies that convey to people
 - ◆ Understand what a product is for and how to use it for an activity.
 - ◆ For example, history and bookmark
 - 2) Concepts to which people are exposed to through the product
 - ◆ ask-domain objects, their attributes, and operations
 - ◆ For example, saving, revisiting, organizing
 - 3) Relationship and mappings between these concepts
 - ◆ Whether one object contains another
 - ◆ Visit a page through history, most-frequently visited, and bookmark
- **Formulating Steps:**
 - 1) 用户任务分析: 确定使用产品时执行的任务
 - 2) 系统支持: 探讨系统如何支持这些任务
 - 3) 界面隐喻和交互模式: 决定哪一种最适合用户理解和使用, 以及这些选择如何影响用户对概念模型的理解

INTERFACE MATAPHORS 界面隐喻

- **Definition:** **Exploit user's familiar knowledge**, helping them to understand 'the unfamiliar'
- **Application Examples:** Conceptualizing what users are doing (surfing the Web); A conceptual model instantiated at the interface (the desktop and card metaphor); Visualizing an operation (an icon of a shopping cart into which the user places items)
- **Benefits:**
 - 1) Makes **learning** new systems easier
 - 2) Helps users understand the underlying conceptual model
 - 3) Can be very **innovative** and enable the realm of computers and their applications to be made **more accessible** to a greater diversity of users
- **Problems:**

- 1) Break conventional and cultural rules (recycle bin placed on desktop)
- 2) Can constrain designers in the way that they conceptualize a problem space
- 3) Conflicts with design principles
- 4) Forces users to understand only the system in terms of the metaphor
- 5) Designers can inadvertently use bad existing designs and transfer the bad parts over
- 6) Limits designers' imagination in coming up with new conceptual models

PARADIGMS 范式 General approach adopted by a community for carrying out research

Shared assumptions, concepts, values, and practices (例如 desktop, ubiquitous computing, in the wild) (Ubiquitous computing / Pervasive computing / Wearable computing / Internet of Things (IoT))

VISIONS 愿景 Provide concrete scenarios of how society can use the next generation of imagined technologies

Also raise ethical questions such as, privacy and trust

THEORIES 理论 Explanation of a phenomenon (information processing that explains how the mind, or some aspect of it, is assumed to work)

MODELS 模型 A **simplification** of an HCI phenomenon

Enables designers to predict and evaluate alternative designs

FRAMEWORKS 框架 **Set of interrelated concepts and-or specific questions for 'what to look for'** (Collaborative learning, ethnographic studies)

Provide advice on how to design user experiences

Focus on how to design particular kinds of interfaces to evoke certain responses (Such as steps, questions, concepts, challenges, principles, tactics, and dimensions)

PROTOTYPING

- **Definition:** instead of an object, it is more a way of thinking, could be [screen sketches, storyboard, PPT, video simulating the system, lump of wood, cardboard mock-up, software with limited functions]
- **Why prototype?**
 - **Understanding:**
 - ◆ Design alternative: make more prototypes and try to understand
 - ◆ Strategy: shows the pros and cons intangible to tangible
 - ◆ User-centered process: avoid false assumptions and build empathy between designers and users with concrete ideas (Empathy map 'say and do', 'see', 'think and feel', 'hear')
 - **Communication:**
 - ◆ Increase efficiency in making decisions with same language
 - ◆ Different stakeholder
 - ◆ No "maybe": helps save efforts in determining style, size, color, and interactions through observable objects and peer discussion
 - **Test and reflection:** to examine and improve the current design
 - ◆ Hypotheses and assumptions: help test the assumptions before design
 - ◆ Others' comments: present and promote the idea to peers and reflect on their comments

FIDELITY

Refers to how it conveys the look-and-feel of the final product. fidelity is a **spectrum**

- 1) **Low fidelity:** useless, quick, cheap, easily change.
- Storyboard,
- Sketching
- Index cards: Often used in website development, application, miniProgram
- Wizard-of-Oz: used in early design to help team know the user's expectation
- 2) **High fidelity:** waste of time and efforts, looks more like final product
 - Uses materials that you expect to be in the final product
 - The consumption of resource and time. Flexibility and iteration difficulties.
 - Compromise (all prototypes involved cannot be ignored)
 - Two types:
 - Horizontal: a wide range of functions but little details
 - Vertical: a lot of details but a few functions
 - Importance reason: Wasting time and sanity with imprecise language about fidelity. Even worse, we risk falsely validating one thing today and failing to validate a better way tomorrow. Teams are prone to over-engineering a prototype prematurely, leading to wasted effort. In other words, a prototype may be low fidelity except for one crucial dimension that may require quality and fidelity at near-production levels to validate fairly.

FIVE DIMENSIONS

- 1. **Visual** 视觉保真度: **most direct dimension** that user feels about the prototype, refers to the devotion and granularity put into the design for **visual interfaces** and **physical objects**
 - a) early stages, rather low visual fidelity, so users will focus on the **actions / procedures / concepts**
 - b) later stages, it should be higher to test the **accessibility / touch / visual detail**
- 2. **Interaction** 交互保真度: Refers to the objective degree of exactness with which real-world interactions can be reproduced (click / tap / switch), allowed the team to quickly and cheaply test hundreds of variations.
 - a) Low interaction: 静态演示
 - b) High interaction: 更多交互元素, 来模拟交互
- 3. **Breadth** 宽度保真度: Refers to **how many feature can be covered**. 例如在商城页面的设计从哪个页面开始, 是否要覆盖产品页、过滤、愿望清单等
- 4. **Depth** 深度保真度: Refers to the degree of functionality
 - a) High depth: 用户可以询问任何与天气相关的问题并得到回答
 - b) Low depth: 用户只能查询当地的温度
- 5. **Content** 内容保真度: Refers to the **degree of realism** for the content. 早期可以低内容保真, 保证功能和流程, 后期要提高
 - a) High: 与实际产品相同的内容
 - b) Low: 使用虚拟或占位符内容
- 6. **Other fidelity**
 - a) Autonomy: operates alone vs requires "supervision"
 - b) Platform: interim (更灵活, 允许快速迭代) **VS** final implementation (更稳定可靠, 直接用于生产环境)

PHYSICAL DESIGN

1. **Definition:** A constant step-by-step process for concerting the product.
2. **Components:**
1. **Menu design:** How long? In what order the items appear? How be structured? What categories? division into groups (color / lines)? How many menus? Terminologies 术语 (climbing example)? Physical constrains?
 2. **Icon design:** culture and context sensitive (draws on existing traditional / concrete objects or things)
 3. **Splitting functions across screens:** moving around within and between screen show much interaction per screen? serial or workbench style? Information available in many screens
 4. **Individual screen design:** white space: balance between enough information/interaction and clarity; grouping items together: separation with boxes? lines? colors? Draw user attention to **salient point**. **Tradeoff** between sparse population and overcrowding
 5. **Information display:** **Relevant** information available at all times; Different types of information imply different kinds of display; Consistency between paper display and screen data entry

PROTOTYPING TYPES

1. **Throw-away:** Prototype only serves to elicit user reaction; Creating prototype must be rapid, otherwise too expensive
2. **Incremental:** Product built as separate components (modules); Each component prototyped and tested, then added to the final system
3. **Evolutionary:** Prototype altered to incorporate design changes; Eventually becomes the final product

CAUTIONS

- Rationalize cost-value tradeoffs
- Do not oversell - capabilities that can't be delivered, development completeness
- Do not overbuild - "good enough" as a prototype
- Decide early on exploratory or evolutionary prototypes

PROCESSES in PROTOTYPE

- Processes start from **minimum variable product**, focusing on exploration (facing different solutions), focusing on **specific audience** (Communicate or advocate for a certain direction), focusing on **assumption** (have a question to test)
- **General Framework:** start from users or audience (再开始设计之前必须了解用户的需求和通电, 通过用户调研、访谈、问卷); iterative process (不断测试和反馈来改进优化产品, 每次迭代更接近产品目标); many paths (在原型设计中采取不同的路径和方法, 选择最合适的) (没有 fix 的方式, 要 based on your final **purpose**, audience, and assumptions 选择路径)
 - **Specific audience – 3 steps:** identity audience, purpose, fidelity; tools; presents
 - **MINIMUM VARIABLE PRODUCT:** A product with **enough features** to **attract** early-adopter customers and **validate** a product idea early in the product development cycle.
 - **4 steps:** Understand your users and identify the problem ; Write down the user flow (详细描述用户在使用产品的每一个步骤和交互点); Use the prototype to optimize the user flow; Test, refine, iterate.
 - **EXPLORATION:** focusing on thinking not doing (find the right issue to solve; find solutions; identify where we are now)
 - **3 steps:** Brainstorming about issues/solutions (团队集思广益, 提出可能的解决方案, 不受限); Clustering and categorization (相似的方案归类, 识别主要方向和问题); Priority check (根据价值和可行性优先级排序, 选择最具潜力的方案深入调查)
 - **ASSUMPTION:** Improve your ideas and products. The assumptions could be either abstract or concrete.
 - **4 steps:** Understand your audience, issues, and assumptions; Fidelity (对于概念、术语、导航、用户识别要低精度; 对于完成度、视觉细节、内容要高精度); Purpose of the testing; Make and test (可以用 A/B test; 制作符合测试的原型, 不需要为所有内容创建 UI 界面; 收集反馈、迭代优化)

DIGITAL and PHYSICAL PROTOTYPING

DIGITAL: features built around screen interaction. 屏幕作为一个空间用 xy 的二维坐标表示, z 表示层次和页面重叠的效果; 响应式设计分为移动端 (不同屏幕尺寸, 不同 UI 元素, 安卓苹果) 和桌面端 (不同浏览器的窗口和大小, 不同屏幕尺寸断点、交互逻辑); 多种交互类型如点击、双击、捏合 pinch、滑动 swipe、翻转 flip、旋转、按压等; 要确保产品对所有用户可用, 涵盖物理、视觉、体验、偏好等方面; 动画 animation 是最佳实现之一 (低保真用绘图实现、高保真用软件实现)

PHYSICAL: focuses more on the → Electronic technology (使用工具包开始, 包括磁铁 magnet、信号 signal、导线 wire、插座 socket) → Compatibility (编码保证可靠性、一致性、避免连接丢失和信号错误的问题) → Material and tactility (用户体验是最直接的, 考虑设备的使用场景、交互方式、使用频率、用户群体和可持续性)

The questions are divided into **closed-ended** & **open-ended**

QUESTIONNAIRE

- **Design:**
 - The design will be affected by question order; Different groups need different questionnaires
 - Need clear guidance; Avoid long questions
 - Decision phrases are all -positive/ all -negative/ mixture
- **Response Format:** -- 'yes/ no' checkboxes, many options checkboxes; rating scales; open-ended response
- **Pros** – low cost for numerous data; need short time; no extra devices; survey are generally approved by institutional review boards because they are typically non-intrusive
- **Cons** -- Shallow data from most people; self-administered (no follow-up questions); caused biased data instead of clear factual phenomena.
- **How to deploy?**
 - Plan timeline
 - Design the questionnaire offline
 - Program/ complete online survey
 - Test to make sure behaves as expect
 - Test with group to check it is clear
 - Recruit participants

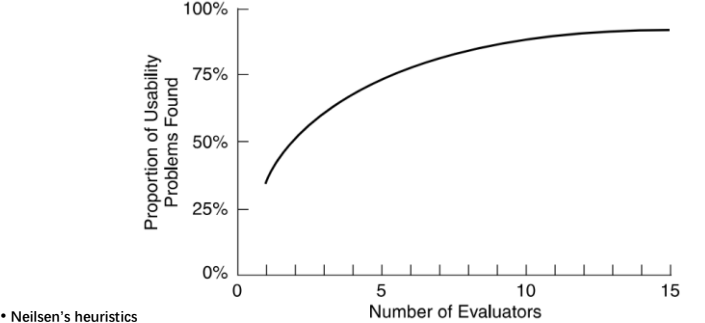
INTERVIEW

- **Types:**
 - Unstructured: not directed by a script, rich but not replicable
 - Structured: tightly scripted, often like a questionnaire, replicable but may lack richness
 - Semi-structured: guided by a script, but interesting issues can be explored in more depth, can provide a

- good balance between richness and replicability.
- **Cautions:** avoid long questions; compound sentences; jargon and language that not be understood; leading questions; unconscious biases
 - **Running the interview**
 - Introduction: introduce yourself, goals, reassure about ethical issue, ask to record, present informed consent form
 - Warm-up: make first question easy and non-threatening
 - Main body: present questions in logical order
 - A cool-off period: include a few easy questions to defuse tensions at the end
 - Closure: thank, signal the end (switch record off)

- **Pros**
 - Go deep: encourage reflection and consideration
 - Flexible: open-ended and exploratory
- **Cons**
 - Skill to manage; time and resource intensive; data analysis; recall problems; separated from task and context under consideration
- **Five key issues of data collection**
 - **Setting goals:** what information to collect, how to analyze collection data
 - **Identifying participants:** decide from whom to gather data and how many
 - **Relationship with participant:** clear and professional, informed consent when appropriate
 - **Triangulation:** look at data from more than one perspective, collect more types data (quantitative data from experiments and qualitative data from interviews)
 - **Pilot studies:** 试点研究

HERUISTIC EVALUATION



- **Neilsen's heuristics**
 - **Visibility of systems status:** design always keep users **informed** about what happen, through appropriate **feedback** within a reasonable amount of time
 - **Match between system and real word:** design should speak **the users' language** (natural logic)
 - **User control and freedom:** an action to leave **unwanted action**
 - **Consistency and standards:** no confusion on different button, situation, actions
 - **Error prevention:** **prevent** error from the first time
 - **Recognition rather than recall:** minimize **memory load** by making options **visible**
 - **Flexibility and efficiency of use:** allow user to **tailor frequent action**
 - **Aesthetic and minimalist design:** not contain **irrelevant** and **rarely needed** information
 - **Help users recognize, diagnose, recover from errors:** error message **indicate** the error and **suggested solution**
 - **Help documentation number of evaluators and problems:** documentation to help users

- **Process of 启发性评估**
 - Briefing session to tell experts what to do
 - Evaluation of period of 1-2 hours in: ①take one pass to get a feel for product; ②take a second pass to focus on specific features
 - **Debriefing session** in which experts work together to prioritize problems
 - ◆ **Advantage:** few ethical and practical issues to consider because users are not involved
 - ◆ **Drawbacks:** difficult and expensive to find experts; important problems may be missed; experts have biases; many trivial problems are often identified (false alarms)

CONGNITIVE WALKTHROUGHS

- Simulating how users go about problem-solving at each step in a human-computer interaction, focus on ease of learning
- **Procedure:**
 1. Preparations
 - ◆ Identify and document the characteristics of typical users.
 - ◆ Develop sample tasks, focusing on the aspects of the design.
 - ◆ Produce a description, mock-up, or prototype of the interface to be developed, along with a clear sequence of the actions needed for the users to complete the task.
 2. A designer and one or more researchers come together to do the analysis.
 3. The researchers walk through the action sequences for each task, placing it within the context of a typical scenario. As they do this, they try to answer three questions.
 4. Compile a record of critical information – assumptions, notes of issues and design changes, etc.
 5. (Check with real users and) Revise the design to fix the problems presented.

WHAT/ WHY/ WHERE/ WHEN to EVALUATE

- **Why:** To **check** users' requirements and **confirm** that users can utilize the product and that they like it
- **What:** A **conceptual model**, early and subsequent **prototypes** of a new system, more **complete prototypes**, and a prototype to compare with **competitors'** products
- **Where:** In **natural**, in-the-wild, and **laboratory settings**
- **When:** **Throughout design**; finished products can be evaluated to collect information to inform new products
- **HCI Research Methods and Measurement (task-based):**

- Time performance: 完成任务所需的时间, 帮助确定效率和响应速度
- Task correctness/ accuracy: 衡量任务能否正确完成, 用于评估可用性和用户体验
- Error rate: 发生的错误数量, 发现缺陷
- Time to learn and retention over time: 学习操作所需要的时间, 评估学习曲线和学习的可持续性
- User satisfaction: 满意度和意见, 帮助提升整体体验和反馈

EVALUATION METHODS

Practical Challenges of Evaluation

1. Agreement of participants: 参与者需要知道为什么进行评估, 被要求做什么, 他们的权利 (Informed consent forms) the evaluation process, data analysis, and data storage methods are typically approved by a high authority, such as the Institutional Review Board
2. Interpreting Data: **Reliability:** Does the method produce the same results on separate occasions? **Validity:** Does the method measure what it is intended to measure? **Ecological validity:** Does the environment of the evaluation distort the results? **Biases:** Are there biases that distort the results? **Scope:** How generalizable are the results?

Evaluation Method Types

1. **Controlled Setting** (observing / asking users / testing)
 - Usability Testing (applied experimentation)
 - 关键要素: **controlled setting:** 测试在受控环境中进行, 保证测试条件的一致性, 控制测试过程 **users are observed and timed:** 评估效率 **recording data:** 用视频, 用于识别问题、改进设计 **performance time and error identification:** 评估易用性和用户体验, 进行改进 **user satisfaction:** 观察、问卷, 评估满意度, 了解需求 **field observing:** 获取用户关于行为和环境的的信息, 提供背景和上下文理解
 - 条件: 在设置好的受控环境中进行, 重点在选择代表性用户和任务; 5-10 名用户 (取决于测试的时间表, 参与者的可用性和测试成本); 持续 30 分钟左右; 测试条件对每个用户都相同; 知情同意书解释程序和伦理问题
 - Experimental (scientific research with relationship hypothesis between variables)
 - 重要信息: **test hypothesis:** 研究人员提出假设变量之间的关系; **relationship of variables:** 自变量有研究人员控制 independent variable, 因变量; **experimental design:** 一个或两个自变量; **Validated statistically and replicable:** 通过统计方法验证, 以确保结果的可靠性, 实验必须有可重复性, 在不同环境时间下有相似结果, 有助于稳健性 robustness 和一致性
2. **Natural Setting** (observing / asking users / asking experts)
 - Field Study
 - ◆ Seek to understand what users do naturally and how technology impacts them
 - ◆ 为了 Identify opportunities for new technology; Determine design requirements; Decide how best to introduce new technology; Evaluate technology in use
 - ◆ 实例: Paipad
3. **Without Users** (testing / modelling)
 - Heuristic Evaluation
 - Analytics
 - 1) **Definition:** The software can automatically record various user actions, such as keystrokes, mouse or other movements pointing to the device, time spent searching on the web, viewing the help system, task flow in the software module, screen time, and movement in the virtual environment.
 - 2) Advantages: 在不影响系统的情况下是不显眼的 unobtrusive; 自动记录大量数据并利用可视化工具
 - 3) Drawbacks: 在不知的情况下观察参与者, 会引发道德和隐私的问题
- A/B Testing: a **large-scale** experiment, compare **two versions** performance, suitable for **evaluating changes in design**, might cause **ethical dilemmas** if users do not know they involved
- Predictive Models: evaluating **without directly involving users** (less expensive than user testing); Use **formulas** to derive various measures of **user performance**; Usefulness limited to systems with **predictable tasks** (voicemail systems, smartphones, and dedicated mobile devices)

4. **Observation**
 - **Types:**
 - ◆ **Direct observation:**
 - In the field:
 - 组成于 who, where, what; 参与程度分为 passive observer 和 active participants; 决定 如何 gain acceptance, 如何解决敏感话题, 如何收集数据 (什么数据、什么装置、什么时候停止数据收集)
 - 民族志 Ethnography: 一种哲学和技术集合, 包括参与观察和访谈, 要求研究者深入了解并参与到研究文中; 在线民族志探讨了在线和线下的研究, 以及虚拟世界的持久性和与物理世界的不同
 - Think-aloud: 一种常用语控制环境中的观察技术, 让用户在执行任务时描述他们的思考过程
 - In controlled environments.
 - ◆ **Indirect observation:** tracking users' activities; Diaries; Interaction logging; Video and photographs collected remotely by drones or other equipment